

Concept 3 | Summation

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$\sum (au_k + bv_k) = a \sum u_k + b \sum v_k$$

Standard sums

$$\sum k = \frac{n(n+1)}{2}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Worked Solutions

Summation

Q001 Written

Write $\sum_{k=1}^5 2k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 2k = 2 + 4 + 6 + 8 + 10$

Final answer

$$2 + 4 + 6 + 8 + 10$$

Q002 Written

Write $\sum_{k=1}^5 (2k + 1)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 (2k + 1) = 3 + 5 + 7 + 9 + 11$

Final answer

$$3 + 5 + 7 + 9 + 11$$

Worked Solutions

Summation

Q003 Written

Write $\sum_{k=1}^5 3k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 3k = 3 + 6 + 9 + 12 + 15$

Final answer

$$3 + 6 + 9 + 12 + 15$$

Q004 Written

Write $\sum_{k=1}^4 (4k - 2)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 (4k - 2) = 2 + 6 + 10 + 14$

Final answer

$$2 + 6 + 10 + 14$$

Worked Solutions

Summation

Q005 Written

Write $\sum_{k=1}^5 (2k - 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 (2k - 3) = -1 + 1 + 3 + 5 + 7$

Final answer

$$-1 + 1 + 3 + 5 + 7$$

Q006 MCQ

Select the correct option: Expand $\sum_{k=1}^3 4k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 8 + 12$$

Worked Solutions

Summation

Q007 MCQ

Select the correct option: Expand $\sum_{k=1}^4 2^k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$2 + 4 + 8 + 16$$

Q008 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (5^k - 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 24 + 124$$

Worked Solutions

Summation

Q009 MCQ

Select the correct option: Expand $\sum_{k=1}^5 (k + 4)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$5 + 6 + 7 + 8 + 9$$

Q010 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (2k + 3)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$7 + 9 + 11 + 13$$

Worked Solutions

Summation

Q011 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (2^k + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 5 + 9$$

Q012 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 3k)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 10 + 18$$

Worked Solutions

Summation

Q013 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (k^2 + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$5 + 10 + 17 + 26$$

Q014 Written

Write $\sum_{k=1}^3 (k^2 - 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (k^2 - 3) = -2 + 1 + 6$

Final answer

$$-2 + 1 + 6$$

Worked Solutions

Summation

Q015 Written

Write $\sum_{k=1}^3 (3k^2 + 2)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (3k^2 + 2) = 5 + 14 + 29$

Final answer

$$5 + 14 + 29$$

Q016 Written

Write $\sum_{k=1}^4 k^2$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 k^2 = 1 + 4 + 9 + 16$

Final answer

$$1 + 4 + 9 + 16$$

Worked Solutions

Summation

Q017 Written

Write $\sum_{k=1}^3 (2k^2 + 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (2k^2 + 3) = 5 + 11 + 21$

Final answer

$$5 + 11 + 21$$

Q018 Written

Write $\sum_{k=1}^4 (k^2 - 2k)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 (k^2 - 2k) = -1 + 0 + 3 + 8$

Final answer

$$-1 + 0 + 3 + 8$$

Worked Solutions

Summation

Q019 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (2k + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 5 + 7 + 9$$

Q020 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (3k - 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$5 + 8 + 11 + 14$$

Worked Solutions

Summation

Q021 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (2k^2 - 3)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$-1 + 5 + 15 + 29$$

Q022 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (3k^2 + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 13 + 28 + 49$$

Worked Solutions

Summation

Q023 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (k^2 - k)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$0 + 2 + 6 + 12$

Q024 MCQ

Select the correct option: Expand $\sum_{k=2}^4 5k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$10 + 15 + 20$

Worked Solutions

Summation

Q025 MCQ

Select the correct option: Expand $\sum_{k=1}^4 3$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 3 + 3 + 3$$

Q026 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (5k + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$6 + 11 + 16 + 21$$

Worked Solutions

Summation

Q027 Written

Write $\sum_{k=1}^4 3^k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 3^k = 3 + 9 + 27 + 81$

Final answer

$$3 + 9 + 27 + 81$$

Q028 Written

Write $\sum_{k=1}^3 2$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^3 2 = 2 + 2 + 2$

Final answer

$$2 + 2 + 2$$

Worked Solutions

Summation

Q029 Written

Write $\sum_{k=1}^4 2^k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 2^k = 2 + 4 + 8 + 16$

Final answer

$$2 + 4 + 8 + 16$$

Q030 Written

Write $\sum_{k=1}^6 3$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^6 3 = 3 + 3 + 3 + 3 + 3 + 3$

Final answer

$$3 + 3 + 3 + 3 + 3 + 3$$

Worked Solutions

Summation

Q031 Written

Write $\sum_{k=1}^3 3^k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 3^k = 3 + 9 + 27$

Final answer

$$3 + 9 + 27$$

Q032 Written

Write $\sum_{k=1}^3 (5^k + 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (5^k + 3) = 8 + 28 + 128$

Final answer

$$8 + 28 + 128$$

Worked Solutions

Summation

Q033 Written

Write $\sum_{k=1}^4 (2^k - 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 (2^k - 3) = -1 + 1 + 5 + 13$

Final answer

$$-1 + 1 + 5 + 13$$

Q034 Written

Write $\sum_{k=1}^4 5$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^4 5 = 5 + 5 + 5 + 5$

Final answer

$$5 + 5 + 5 + 5$$

Worked Solutions

Summation

Q035 Written

Write $\sum_{k=1}^3 3$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^3 3 = 3 + 3 + 3$

Final answer

$$3 + 3 + 3$$

Q036 Written

Write $\sum_{k=1}^5 4$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^5 4 = 4 + 4 + 4 + 4 + 4$

Final answer

$$4 + 4 + 4 + 4 + 4$$

Worked Solutions

Summation

Q037 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 2)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 6 + 11$$

Q038 MCQ

Select the correct option: Expand $\sum_{k=1}^5 7$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$7 + 7 + 7 + 7 + 7$$

Worked Solutions

Summation

Q039 MCQ

Select the correct option: Expand $\sum_{k=3}^6 k^2$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$9 + 16 + 25 + 36$$

Q040 MCQ

Select the correct option: Expand $\sum_{k=1}^3 6^k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$6 + 36 + 216$$

Worked Solutions

Summation

Q041 MCQ

Select the correct option: Expand $\sum_{k=1}^5 (4k - 3)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$1 + 5 + 9 + 13 + 17$$

Q042 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (3k - 5)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$-2 + 1 + 4 + 7$$

Worked Solutions

Summation

Q043 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (4^k - 2)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$2 + 14 + 62$$

Q044 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (2k^2 - k)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$1 + 6 + 15$$

Worked Solutions

Summation

Q045 Written

Express $3 + 7 + 11 + \cdots + (4n - 1)$ **using the summation symbol** Σ .**Solution**

1. The general term is $4k - 1$.
2. $\sum_{k=1}^n (4k - 1)$

Final answer

$$\sum_{k=1}^n (4k - 1)$$

Q046 Written

Express $5 + 8 + 11 + \cdots + (3n + 2)$ **using** Σ .**Solution**

1. The general term is $3k + 2$.
2. $\sum_{k=1}^n (3k + 2)$

Final answer

$$\sum_{k=1}^n (3k + 2)$$

Worked Solutions

Summation

Q047 Written

Write the sum of the first n terms of the sequence $a_n = 2n + 1$ using Σ .

Solution

1. Change the term index from n to k .
2. $\sum_{k=1}^n (2k + 1)$

Final answer

$$\sum_{k=1}^n (2k + 1)$$

Q048 Written

Express $1 + 3 + 5 + \dots + (2n - 1)$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (2k - 1)$

Final answer

$$\sum_{k=1}^n (2k - 1)$$

Worked Solutions

Summation

Q049 Written

Write the sum of the first n terms of $a_n = 7n - 1$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (7k - 1)$

Final answer

$$\sum_{k=1}^n (7k - 1)$$

Q050 Written

Write the sum of the first n terms of $a_n = 4n + 5$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (4k + 5)$

Final answer

$$\sum_{k=1}^n (4k + 5)$$

Worked Solutions

Summation

Q051 Written

Write the sum of the first n terms of $a_n = 3n - 2$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (3k - 2)$

Final answer

$$\sum_{k=1}^n (3k - 2)$$

Q052 MCQ

Select the correct option: Express $2 + 5 + 8 + \cdots + (3n - 1)$ using the summation symbol Σ :

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n (3k - 1)$$

Worked Solutions

Summation

Q053 MCQ

Select the correct option: Express $7 + 11 + 15 + \dots + 43$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=1}^{10} (4k + 3)$$

Q054 MCQ

Select the correct option: Express $6 + 10 + 14 + \dots + (4n + 2)$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=1}^n (4k + 2)$$

Worked Solutions

Summation

Q055 MCQ

Select the correct option: Express $8 + 13 + 18 + \dots + 103$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^{20} (5k + 3)$$

Q056 MCQ

Select the correct option: Express $11^2 + 12^2 + \dots + 19^2$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=11}^{19} k^2$$

Worked Solutions

Summation

Q057 MCQ

Select the correct option: Express $2^2 + 3^2 + \dots + 10^2$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=2}^{10} k^2$$

Q058 MCQ

Select the correct option: Express $1 + 3 + 9 + \dots + 3^{n-1}$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=0}^{n-1} 3^k$$

Worked Solutions

Summation

Q059 MCQ

Select the correct option: Express $2 + 7 + 12 + \dots + (5n - 3)$ using the summation symbol Σ :

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n (5k - 3)$$

Q060 Written

Express $6^2 + 7^2 + 8^2 + \dots + 15^2$ using Σ .

Solution

1. The index starts at 6 and ends at 15.

2. $\sum_{k=6}^{15} k^2$

Final answer

$$\sum_{k=6}^{15} k^2$$

Worked Solutions

Summation

Q061 Written

Express $4^2 + 5^2 + 6^2 + \cdots + 13^2$ **using** Σ .**Solution**

1. The index starts at 4 and ends at 13.
2. $\sum_{k=4}^{13} k^2$

Final answer

$$\sum_{k=4}^{13} k^2$$

Q062 Written

Express $1 + 7 + 25 + 79 + \cdots + (3^n - 2)$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (3^k - 2)$

Final answer

$$\sum_{k=1}^n (3^k - 2)$$

Worked Solutions

Summation

Q063 Written

Express $1 + 2 + 4 + 8 + \dots + 2^{n-1}$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n 2^{k-1}$

Final answer

$$\sum_{k=1}^n 2^{k-1}$$

Q064 Written

Express $7^2 + 8^2 + \dots + 21^2$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=7}^{21} k^2$

Final answer

$$\sum_{k=7}^{21} k^2$$

Worked Solutions

Summation

Q065 Written

Express $12^2 + 13^2 + \dots + 18^2$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=12}^{18} k^2$

Final answer

$$\sum_{k=12}^{18} k^2$$

Q066 Written

Express $10^2 + 11^2 + \dots + 20^2$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=10}^{20} k^2$

Final answer

$$\sum_{k=10}^{20} k^2$$

Worked Solutions

Summation

Q067 MCQ

Select the correct option: Express $4^2 + 5^2 + \dots + 12^2$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=4}^{12} k^2$$

Q068 MCQ

Select the correct option: Express $3^1 + 3^2 + \dots + 3^n$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n 3^k$$

Worked Solutions

Summation

Q069 MCQ

Select the correct option: Express $2 + 4 + 8 + \dots + 2^n$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=1}^n 2^k$$

Q070 MCQ

Select the correct option: Express $1 + 2^1 + 2^2 + \dots + 2^{n-1}$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=0}^{n-1} 2^k$$

Worked Solutions

Summation

Q071 MCQ

Select the correct option: Express $7 + 14 + 21 + \dots + 7n$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n 7k$$

Q072 MCQ

Select the correct option: Express $10 + 15 + 20 + \dots + 5n$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=2}^n 5k$$

Worked Solutions

Summation

Q073 MCQ

Select the correct option: Express $6^2 + 7^2 + \dots + 16^2$ using the summation symbol Σ :

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=6}^{16} k^2$$

Q074 Written

Express $2 + 5 + 8 + \dots + 29$ using Σ .

Solution

1. The general term is $3k - 1$, and $29 = 3(10) - 1$.
2. $\sum_{k=1}^{10} (3k - 1)$

Final answer

$$\sum_{k=1}^{10} (3k - 1)$$

Worked Solutions

Summation

Q075 **Written****Express** $8 + 16 + 24 + \cdots + 56$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^7 8k$

Final answer

$$\sum_{k=1}^7 8k$$

Q076 **Written****Express** $7 + 13 + 19 + \cdots + 73$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{12} (6k + 1)$

Final answer

$$\sum_{k=1}^{12} (6k + 1)$$

Worked Solutions

Summation

Q077 Written

Express $4 + 9 + 14 + \dots + 119$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{24} (5k - 1)$

Final answer

$$\sum_{k=1}^{24} (5k - 1)$$

Worked Solutions

Summation

Q078 Challenge

A green-tech startup installs 1 solar panel in Week 1, 4 in Week 2, 7 in Week 3, and $3n - 2$ in Week n . Write one expression using Σ for the total panels installed from Week 1 through Week n .

Solution

1. The weekly pattern is arithmetic with general term $3k - 2$.
2. Total = $\sum_{k=1}^n (3k - 2)$

Final answer

$$\sum_{k=1}^n (3k - 2)$$

Worked Solutions

Summation

Q079 Written

Find $\sum_{k=1}^n (-3)$.

Solution

1. There are n copies of -3 .
2. $\sum_{k=1}^n (-3) = -3n$

Final answer $-3n$

Q080 Written

Find $\sum_{k=1}^n 3$.

Solution

1. There are n copies of 3.
2. $\sum_{k=1}^n 3 = 3n$

Final answer $3n$

Worked Solutions

Summation

Q081 Written

Find $\sum_{k=1}^n 6$.

Solution

1. $\sum_{k=1}^n 6 = 6n$

Final answer

$6n$

Q082 Written

Find $\sum_{k=1}^n (-4)$.

Solution

1. $\sum_{k=1}^n (-4) = -4n$

Final answer

$-4n$

Worked Solutions

Summation

Q083 Written

Find $\sum_{k=1}^n 1$.

Solution

1. $\sum_{k=1}^n 1 = n$

Final answer

 n

Q084 MCQ

Select the correct option: Find $\sum_{k=1}^n 9$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

 $9n$

Worked Solutions

Summation

Q085 MCQ

Select the correct option: Find $\sum_{k=1}^n (-2)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer $-2n$

Q086 MCQ

Select the correct option: Find $\sum_{k=1}^7 4$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer 28

Worked Solutions

Summation

Q087 MCQ

Select the correct option: Find $\sum_{k=1}^{12} (-5)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

−60

Q088 MCQ

Select the correct option: Find $\sum_{k=1}^n k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$\frac{n(n+1)}{2}$

Worked Solutions

Summation

Q089 MCQ

Select the correct option: Find $\sum_{k=1}^n 4k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$2n(n + 1)$$

Q090 MCQ

Select the correct option: Find $\sum_{k=1}^{10} k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$55$$

Worked Solutions

Summation

Q091 MCQ

Select the correct option: Find $\sum_{k=1}^8 3k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

108

Q092 MCQ

Select the correct option: Find $\sum_{k=1}^6 5k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

105

Worked Solutions

Summation

Q093 MCQ

Select the correct option: Find $\sum_{k=1}^{20} k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

210

Q094 MCQ

Select the correct option: Find $\sum_{k=1}^n k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer
$$\frac{n(n+1)(2n+1)}{6}$$

Worked Solutions

Summation

Q095 MCQ

Select the correct option: Find $\sum_{k=1}^n 6k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$n(n+1)(2n+1)$$

Q096 MCQ

Select the correct option: Find $\sum_{k=1}^5 k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$55$$

Worked Solutions

Summation

Q097 MCQ

Select the correct option: Find $\sum_{k=1}^4 2k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

60

Q098 MCQ

Select the correct option: Find $\sum_{k=1}^3 4k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

56

Worked Solutions

Summation

Q099 MCQ

Select the correct option: Find $\sum_{k=1}^7 k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

140

Q100 Written

Find $\sum_{k=1}^n (5k + 4)$.

Solution

1. $\sum (5k + 4) = 5 \sum k + \sum 4$

2. $= \frac{5n(n+1)}{2} + 4n = \frac{n(5n+13)}{2}$

Final answer

$$\frac{n(5n + 13)}{2}$$

Worked Solutions

Summation

Q101 Written

Find $\sum_{k=1}^n (2k - 3)$.

Solution

$$1. \ 2 \sum k - \sum 3 = n(n+1) - 3n$$

$$2. \ = n(n-2)$$

Final answer

$$n(n-2)$$

Q102 Written

Find $\sum_{k=1}^n (4k - 1)$.

Solution

$$1. \ 4 \sum k - \sum 1 = 2n(n+1) - n$$

$$2. \ = n(2n+1)$$

Final answer

$$n(2n+1)$$

Worked Solutions

Summation

Q103 Written

Find $\sum_{k=1}^n (3k + 2)$.

Solution

$$1. \quad 3 \sum k + \sum 2 = \frac{3n(n+1)}{2} + 2n$$

$$2. \quad = \frac{n(3n+7)}{2}$$

Final answer

$$\frac{n(3n+7)}{2}$$

Q104 Written

Find $\sum_{k=1}^n (6k - 3)$.

Solution

$$1. \quad 6 \sum k - \sum 3 = 3n(n+1) - 3n$$

$$2. \quad = 3n^2$$

Final answer

$$3n^2$$

Worked Solutions

Summation

Q105 MCQ

Select the correct option: Find $\sum_{k=1}^n (3k - 1)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(3n + 1)}{2}$$

Q106 MCQ

Select the correct option: Find $\sum_{k=1}^n (2k + 5)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$n(n + 6)$$

Worked Solutions

Summation

Q107 MCQ

Select the correct option: Find $\sum_{k=1}^n (8k - 4)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$4n^2$$

Q108 MCQ

Select the correct option: Find $\sum_{k=1}^n (5k + 3)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(5n + 11)}{2}$$

Worked Solutions

Summation

Q109 Written

Find $\sum_{k=1}^n (k-1)^2$.

Solution

$$1. \sum (k^2 - 2k + 1) = \sum k^2 - 2 \sum k + \sum 1$$

$$2. = \frac{n(n-1)(2n-1)}{6}$$

Final answer

$$\frac{n(n-1)(2n-1)}{6}$$

Q110 MCQ

Select the correct option: Find $\sum_{k=1}^n (k^2 + k)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Summation

Q111 MCQ

Select the correct option: Find $\sum_{k=1}^n (2k^2 - k)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(4n-1)}{6}$$

Q112 MCQ

Select the correct option: Find $\sum_{k=1}^n (k^2 - 1)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(2n^2 + 3n - 5)}{6}$$

Worked Solutions

Summation

Q113 MCQ

Select the correct option: Find $\sum_{k=1}^n (3k^2 + 2k)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(2n+3)}{2}$$

Q114 MCQ

Select the correct option: Find $\sum_{k=1}^n (4k^2 - 3k + 2)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(8n^2 + 3n + 7)}{6}$$

Worked Solutions

Summation

Q115 Written

Find $1 \cdot 3 + 2 \cdot 8 + 3 \cdot 13 + \cdots + n(5n - 2)$.

Solution

$$\begin{aligned} 1. \quad & \sum_{k=1}^n k(5k - 2) = 5 \sum k^2 - 2 \sum k \\ 2. \quad & = \frac{5n(n+1)(2n+1)}{6} - n(n+1) = \frac{n(n+1)(10n-1)}{6} \end{aligned}$$

Final answer

$$\frac{n(n+1)(10n-1)}{6}$$

Q116 Written

Find $\sum_{k=1}^n (k-1)(3k-1)$.

Solution

$$\begin{aligned} 1. \quad & (k-1)(3k-1) = 3k^2 - 4k + 1 \\ 2. \quad & 3 \sum k^2 - 4 \sum k + \sum 1 = \frac{n(n-1)(2n+1)}{2} \end{aligned}$$

Final answer

$$\frac{n(n-1)(2n+1)}{2}$$

Worked Solutions

Summation

Q117 Written

Find $1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 + \cdots + n(2n + 1)$.

Solution

$$1. \sum_{k=1}^n k(2k + 1) = 2 \sum k^2 + \sum k$$

$$2. = \frac{n(n+1)(4n+5)}{6}$$

Final answer

$$\frac{n(n+1)(4n+5)}{6}$$

Q118 Written

Find $\sum_{k=1}^n k(k - 2)$.

Solution

$$1. \sum (k^2 - 2k) = \sum k^2 - 2 \sum k$$

$$2. = \frac{n(n+1)(2n-5)}{6}$$

Final answer

$$\frac{n(n+1)(2n-5)}{6}$$

Worked Solutions

Summation

Q119 Written

Find $\sum_{k=1}^n k(2k - 2)$.

Solution

$$1. 2 \sum k^2 - 2 \sum k$$

$$2. = \frac{2n(n+1)(n-1)}{3}$$

Final answer

$$\frac{2n(n+1)(n-1)}{3}$$

Q120 Written

Find $1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 + \cdots + n(n+3)$.

Solution

$$1. \sum_{k=1}^n k(k+3) = \sum k^2 + 3 \sum k$$

$$2. = \frac{n(n+1)(n+5)}{3}$$

Final answer

$$\frac{n(n+1)(n+5)}{3}$$

Worked Solutions

Summation

Q121 Written

Find $1 \cdot 5 + 2 \cdot 8 + 3 \cdot 11 + \cdots + n(3n + 2)$.**Solution**

$$1. \sum_{k=1}^n k(3k + 2) = 3 \sum k^2 + 2 \sum k$$

$$2. = \frac{n(n+1)(2n+3)}{2}$$

Final answer

$$\frac{n(n+1)(2n+3)}{2}$$

Q122 Written

Find $2 \cdot 5 + 3 \cdot 6 + 4 \cdot 7 + \cdots + (n+1)(n+4)$.**Solution**

$$1. \sum_{k=1}^n (k+1)(k+4) = \sum (k^2 + 5k + 4)$$

$$2. = \frac{n(n+4)(n+5)}{3}$$

Final answer

$$\frac{n(n+4)(n+5)}{3}$$

Worked Solutions

Summation

Q123 Written

Find $\sum_{k=1}^4 (2k - 1)$.

Solution

$$1. \ 2 \sum_{k=1}^4 k - \sum_{k=1}^4 1 = 2(10) - 4$$

$$2. \ = 16$$

Final answer

16

Q124 Written

Find $\sum_{k=1}^5 (3k - 2)$.

Solution

$$1. \ 3 \sum_{k=1}^5 k - 2 \sum_{k=1}^5 1 = 3(15) - 10$$

$$2. \ = 35$$

Final answer

35

Worked Solutions

Summation

Q125 Written

Find $\sum_{k=1}^{20} (2k - 7)$.

Solution

1. $2 \sum_{k=1}^{20} k - 7 \sum_{k=1}^{20} 1 = 2(210) - 140$

2. $= 280$

Final answer

280

Q126 Written

Find $\sum_{k=1}^8 (4k + 5)$.

Solution

1. $4 \sum_{k=1}^8 k + 5 \sum_{k=1}^8 1 = 4(36) + 40$

2. $= 184$

Final answer

184

Worked Solutions

Summation

Q127 Written

Find $\sum_{k=1}^{10} (2k - 1)$.

Solution

$$\begin{aligned} 1. & \quad 2 \sum_{k=1}^{10} k - \sum_{k=1}^{10} 1 = 2(55) - 10 \\ 2. & \quad = 100 \end{aligned}$$

Final answer

100

Worked Solutions

Summation

Q128 Challenge

Find $\sum_{r=4}^8 (r^2 + r + 5)$.

Solution

$$\begin{aligned} 1. \quad \sum_{r=4}^8 (r^2 + r + 5) &= \sum_{r=1}^8 (r^2 + r + 5) - \sum_{r=1}^3 (r^2 + r + 5) \\ 2. \quad &= (204 + 36 + 40) - (14 + 6 + 15) = 245 \end{aligned}$$

Final answer

245

Worked Solutions

Summation

Q129 Written

Find the sum $\sum_{k=1}^n k^2(k+1)$.

Solution

1. Expand: $k^2(k+1) = k^3 + k^2$.
2. Use $\sum k^3 = \frac{n^2(n+1)^2}{4}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $\sum_{k=1}^n k^2(k+1) = \frac{1}{12}n(n+1)(n+2)(3n+1)$

Final answer

$$\frac{n(n+1)(n+2)(3n+1)}{12}$$

Q130 Written

Find the sum $\sum_{k=1}^n k^2(k-2)$.

Solution

1. Expand: $k^2(k-2) = k^3 - 2k^2$.
2. Substitute the formulae for $\sum k^3$ and $\sum k^2$, then factor.
3. $\sum_{k=1}^n k^2(k-2) = \frac{n(n+1)(3n^2-5n-4)}{12}$

Final answer

$$\frac{n(n+1)(3n^2-5n-4)}{12}$$

Worked Solutions

Summation

Q131 Written

Find the sum $\sum_{k=1}^n k^2(k+2)$.

Solution

1. Expand: $k^2(k+2) = k^3 + 2k^2$.
2. Substitute the standard sums and factor the common $n(n+1)$.
3. $\sum_{k=1}^n k^2(k+2) = \frac{n(n+1)(3n^2 + 11n + 4)}{12}$

Final answer

$$\frac{n(n+1)(3n^2 + 11n + 4)}{12}$$

Q132 Written

Find the sum $\sum_{k=1}^n k^2(k-3)$.

Solution

1. Expand: $k^2(k-3) = k^3 - 3k^2$.
2. Substitute the standard sums and simplify the common factor.
3. $\sum_{k=1}^n k^2(k-3) = \frac{n(n+1)(n^2-3n-2)}{4}$

Final answer

$$\frac{n(n+1)(n^2-3n-2)}{4}$$

Worked Solutions

Summation

Q133 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+1)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n+2)(3n+1)}{12}$$

Q134 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-5n-4)}{12}$$

Worked Solutions

Summation

Q135 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+11n+4)}{12}$$

Q136 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2-3n-2)}{4}$$

Worked Solutions

Summation

Q137 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+5n+2)}{4}$$

Q138 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-4)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-13n-8)}{12}$$

Worked Solutions

Summation

Q139 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+4)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+19n+8)}{12}$$

Q140 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-1)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-5n-4)}{12}$$

Worked Solutions

Summation

Q141 Written

Find the sum $\sum_{k=1}^n k(k^2 - 1)$.

Solution

1. Expand: $k(k^2 - 1) = k^3 - k$.
2. Use $\sum k^3$ and $\sum k$, then factor $n^2 + n - 2 = (n - 1)(n + 2)$.
3. $\sum_{k=1}^n k(k^2 - 1) = \frac{n(n + 1)(n - 1)(n + 2)}{4}$

Final answer

$$\frac{n(n + 1)(n - 1)(n + 2)}{4}$$

Q142 Written

Find the sum $\sum_{k=1}^n k(k^2 + 2)$.

Solution

1. Expand: $k(k^2 + 2) = k^3 + 2k$.
2. Use $\sum k^3$ and $\sum k$, then factor.
3. $\sum_{k=1}^n k(k^2 + 2) = \frac{n(n+1)(n^2 + n + 4)}{4}$

Final answer

$$\frac{n(n+1)(n^2 + n + 4)}{4}$$

Worked Solutions

Summation

Q143 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(n-1)(n+2)}{4}$$

Q144 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 2)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(n^2 + n + 4)}{4}$$

Worked Solutions

Summation

Q145 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+6)}{4}$$

Q146 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n-4)}{4}$$

Worked Solutions

Summation

Q147 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 4)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+8)}{4}$$

Q148 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n-6)}{4}$$

Worked Solutions

Summation

Q149 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+23n+10)}{12}$$

Q150 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-17n-10)}{12}$$

Worked Solutions

Summation

Q151 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+10)}{4}$$

Q152 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n-10)}{4}$$

Worked Solutions

Summation

Q153 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+6)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2 + 27n + 12)}{12}$$

Q154 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 6)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2 + n + 12)}{4}$$

Worked Solutions

Summation

Q155 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k - 6)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(3n^2 - 25n - 12)}{12}$$

Q156 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 6)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(n^2 + n - 12)}{4}$$

Worked Solutions

Summation

Q157 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+7)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+31n+14)}{12}$$

Q158 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+8)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+35n+16)}{12}$$

Worked Solutions

Summation

Q159 Written

Find the sum $\sum_{k=1}^n 6 \times 3^{k-1}$.

Solution

1. Use $a = 6, r = 3$.
2. $S_n = \frac{6(3^n - 1)}{3 - 1} = 3(3^n - 1)$

Final answer

$$3(3^n - 1)$$

Q160 Written

Find the sum $\sum_{k=1}^n 4 \times 2^{k-1}$.

Solution

1. Use $a = 4, r = 2$.
2. $S_n = \frac{4(2^n - 1)}{2 - 1} = 4(2^n - 1)$

Final answer

$$4(2^n - 1)$$

Worked Solutions

Summation

Q161 Written

Find the sum $\sum_{k=1}^n 2^k$.

Solution

1. Rewrite as $(2 \times 2^{k-1})$.
2. $S_n = \frac{2(2^n - 1)}{2 - 1} = 2(2^n - 1)$

Final answer

$$2(2^n - 1)$$

Q162 Written

Find the sum $\sum_{k=1}^n 2 \times (-3)^{k-1}$.

Solution

1. Here $a = 2$ and $r = -3$.
2. $S_n = \frac{2(1 - (-3)^n)}{1 - (-3)} = \frac{1 - (-3)^n}{2}$

Final answer

$$\frac{1 - (-3)^n}{2}$$

Worked Solutions

Summation

Q163 Written

Find the sum $\sum_{k=1}^n 3 \times (-2)^{k-1}$.

Solution

1. Use $a = 3, r = -2$.
2. $S_n = \frac{3(1-(-2)^n)}{1-(-2)} = 1 - (-2)^n$

Final answer

$$1 - (-2)^n$$

Q164 Written

Find the sum $\sum_{k=1}^n (-5)^k$.

Solution

1. Rewrite as $(-5) \times (-5)^{k-1}$, so $a = -5, r = -5$.
2. $S_n = \frac{-5(1-(-5)^n)}{1-(-5)} = \frac{5((-5)^n - 1)}{6}$

Final answer

$$\frac{5((-5)^n - 1)}{6}$$

Worked Solutions

Summation

Q165 Written

Find the sum $\sum_{k=1}^n 2 \times (-1)^{k-1}$.

Solution

1. Use $a = 2, r = -1$.
2. $S_n = \frac{2(1-(-1)^n)}{2} = 1 - (-1)^n$

Final answer

$$1 - (-1)^n$$

Q166 Written

Find the sum $\sum_{k=1}^n 8 \times (-3)^{k-1}$.

Solution

1. Use $a = 8, r = -3$.
2. $S_n = \frac{8(1-(-3)^n)}{4} = 2(1 - (-3)^n)$

Final answer

$$2(1 - (-3)^n)$$

Worked Solutions

Summation

Q167 Written

Find the sum $\sum_{k=1}^n (-2)^{k-1}$.

Solution

1. Use $(a = 1, r = -2)$.
2. $S_n = \frac{1 - (-2)^n}{1 - (-2)} = \frac{1 - (-2)^n}{3}$

Final answer

$$\frac{1 - (-2)^n}{3}$$

Q168 Written

Find the sum $\sum_{k=1}^n (-3)^k$.

Solution

1. Rewrite as $((-3)(-3)^{k-1})$.
2. $S_n = \frac{-3(1 - (-3)^n)}{4} = \frac{3((-3)^n - 1)}{4}$

Final answer

$$\frac{3((-3)^n - 1)}{4}$$

Worked Solutions

Summation

Q169 Written

Find the sum $\sum_{k=1}^n 4^k$.

Solution

1. Rewrite $4^k = 4 \times 4^{k-1}$, so $a = 4, r = 4$.

2. $S_n = \frac{4(4^n - 1)}{4 - 1} = \frac{4(4^n - 1)}{3}$

Final answer

$$\frac{4(4^n - 1)}{3}$$

Q170 Written

Find the sum $\sum_{k=1}^n 4^{k-1}$.

Solution

1. The first term is 1 and the common ratio is 4.

2. $S_n = \frac{4^n - 1}{4 - 1} = \frac{4^n - 1}{3}$

Final answer

$$\frac{4^n - 1}{3}$$

Worked Solutions

Summation

Q171 Written

Find the sum $\sum_{k=1}^n 3^{k-1}$.

Solution

1. The first term is 1 and the ratio is 3.
2. $S_n = \frac{3^n - 1}{3 - 1} = \frac{3^n - 1}{2}$

Final answer

$$\frac{3^n - 1}{2}$$

Q172 Written

Find the sum $\sum_{k=1}^4 (-2)^{k-1}$.

Solution

1. This is a finite geometric sum with $(a = 1, r = -2, n = 4)$.
2. $S_4 = 1 - 2 + 4 - 8 = -5$

Final answer

$$-5$$

Worked Solutions

Summation

Q173 Written

Find the sum $\sum_{k=1}^3 3^k$.

Solution

1. Use $(a = 3, r = 3, n = 3)$.
2. $S_3 = 3 + 9 + 27 = 39$

Final answer

39

Q174 Written

Find the sum $\sum_{k=1}^5 2^k$.

Solution

1. Use $(a = 2, r = 2, n = 5)$.
2. $S_5 = 2 + 4 + 8 + 16 + 32 = 62$

Final answer

62

Worked Solutions

Summation

Q175 Written

Find the sum $\sum_{k=1}^4 (-3)^k$.

Solution

1. Use $(a = -3, r = -3, n = 4)$.
2. $S_4 = -3 + 9 - 27 + 81 = 60$

Final answer

60

Worked Solutions

Summation

Q176 Challenge

Find the finite sum $\sum_{r=5}^9 (2r + 7)$.

Solution

1. Substitute ($r = 5, 6, 7, 8, 9$): ($17+19+21+23+25$).
2. $17 + 19 + 21 + 23 + 25 = 105$

Final answer

105

Worked Solutions

Summation

Q177 Written

Find $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \cdots + \frac{1}{(n+1)(n+2)}.$

Solution

1. Split each term as $\frac{1}{j} - \frac{1}{j+1}.$
2. $\frac{1}{2} - \frac{1}{n+2} = \frac{n}{2(n+2)}$

Final answer

$$\frac{n}{2(n+2)}$$

Q178 Written

Find $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \cdots + \frac{1}{n(n+1)}.$

Solution

1. Use $\frac{1}{j(j+1)} = \frac{1}{j} - \frac{1}{j+1}.$
2. $1 - \frac{1}{n+1} = \frac{n}{n+1}$

Final answer

$$\frac{n}{n+1}$$

Worked Solutions

Summation

Q179 Written

Find $\frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \cdots + \frac{1}{(n+2)(n+3)}$.

Solution

1. The gap is 1 and the first term starts at 3.

2. $\frac{1}{3} - \frac{1}{n+3} = \frac{n}{3(n+3)}$

Final answer

$$\frac{n}{3(n+3)}$$

Q180 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 2} + \cdots + \frac{1}{n(n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{n+1}$$

Worked Solutions

Summation

Q181 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 3} + \cdots + \frac{1}{(n+1)(n+2)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2(n+2)}$$

Q182 MCQ

Select the correct option: Find the sum $\frac{1}{3 \times 4} + \cdots + \frac{1}{(n+2)(n+3)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{3(n+3)}$$

Worked Solutions

Summation

Q183 MCQ

Select the correct option: Find the sum $\frac{1}{4 \times 5} + \cdots + \frac{1}{(n+3)(n+4)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{4(n+4)}$$

Q184 MCQ

Select the correct option: Find the sum $\frac{1}{6 \times 7} + \cdots + \frac{1}{(n+5)(n+6)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{6(n+6)}$$

Worked Solutions

Summation

Q185 MCQ

Select the correct option: Find the sum $\frac{1}{7 \times 8} + \cdots + \frac{1}{(n+6)(n+7)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{7(n+7)}$$

Q186 MCQ

Select the correct option: Find the sum $\frac{1}{9 \times 10} + \cdots + \frac{1}{(n+8)(n+9)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{9(n+9)}$$

Worked Solutions

Summation

Q187 MCQ

Select the correct option: Find the sum $\frac{1}{10 \times 11} + \cdots + \frac{1}{(n+9)(n+10)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{10(n+10)}$$

Q188 MCQ

Select the correct option: Find the sum $\frac{1}{11 \times 12} + \cdots + \frac{1}{(n+10)(n+11)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{11(n+11)}$$

Worked Solutions

Summation

Q189 Written

Find the following sum: $\frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \cdots + \frac{1}{(3n-1)(3n+2)}$.

Solution

1. Use $\frac{1}{ab} = \frac{1}{b-a} \left(\frac{1}{a} - \frac{1}{b} \right)$ with gap 3.
2. $\frac{1}{3} \left(\frac{1}{2} - \frac{1}{3n+2} \right) = \frac{n}{2(3n+2)}$

Final answer

$$\frac{n}{2(3n+2)}$$

Q190 Written

Find $\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \cdots + \frac{1}{(2n-1)(2n+1)}$.

Solution

1. The gap is 2, so factor out $\frac{1}{2}$.
2. $\frac{1}{2} \left(1 - \frac{1}{2n+1} \right) = \frac{n}{2n+1}$

Final answer

$$\frac{n}{2n+1}$$

Worked Solutions

Summation

Q191 Written

Find $\frac{1}{1 \times 3} + \frac{1}{2 \times 4} + \frac{1}{3 \times 5} + \cdots + \frac{1}{n(n+2)}.$

Solution

1. Use the constant gap 2 decomposition; two end terms remain.
2. $\frac{1}{2}(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2})$

Final answer

$$\frac{1}{2}(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2})$$

Q192 Written

Find $\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \cdots + \frac{1}{(3n-2)(3n+1)}.$

Solution

1. Use gap 3 and the first denominator 1.
2. $\frac{1}{3}(1 - \frac{1}{3n+1}) = \frac{n}{3n+1}$

Final answer

$$\frac{n}{3n+1}$$

Worked Solutions

Summation

Q193 Written

Find $\frac{1}{1 \times 7} + \frac{1}{4 \times 10} + \frac{1}{7 \times 13} + \cdots + \frac{1}{(3n-2)(3n+4)}$.

Solution

1. The two factors differ by 6, while the first factors advance by 3; two terms remain at each end.
2. $\frac{1}{6}(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4})$

Final answer

$$\frac{1}{6}(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4})$$

Q194 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 3} + \cdots + \frac{1}{(2n-1)(2n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2n+1}$$

Worked Solutions

Summation

Q195 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 4} + \cdots + \frac{1}{(3n-2)(3n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{3n+1}$$

Q196 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 7} + \cdots + \frac{1}{(3n-2)(3n+4)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{1}{6} \left(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4} \right)$$

Worked Solutions

Summation

Q197 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 6} + \cdots + \frac{1}{(4n-2)(4n+2)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2(4n+2)}$$

Q198 MCQ

Select the correct option: Find the sum $\frac{1}{3 \times 8} + \cdots + \frac{1}{(5n-2)(5n+3)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{3(5n+3)}$$

Worked Solutions

Summation

Q199 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 5} + \cdots + \frac{1}{(4n-3)(4n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{4n+1}$$

Q200 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 7} + \cdots + \frac{1}{(5n-3)(5n+2)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2(5n+2)}$$

Worked Solutions

Summation

Q201 MCQ

Select the correct option: Find the sum $\frac{1}{4 \times 10} + \cdots + \frac{1}{(6n-2)(6n+4)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{4(6n+4)}$$

Q202 Written

Find $\frac{1}{4 \times 6} + \frac{1}{5 \times 7} + \frac{1}{6 \times 8} + \cdots + \frac{1}{(n+3)(n+5)}$.

Solution

1. Write the general term as $\frac{1}{(j+3)(j+5)}$, gap 2.

2. $\frac{1}{2} \left(\frac{1}{4} + \frac{1}{5} - \frac{1}{n+4} - \frac{1}{n+5} \right)$

Final answer

$$\frac{1}{2} \left(\frac{1}{4} + \frac{1}{5} - \frac{1}{n+4} - \frac{1}{n+5} \right)$$

Worked Solutions

Summation

Q203 Written

Find $\frac{1}{2 \times 4} + \frac{1}{4 \times 6} + \frac{1}{6 \times 8} + \cdots + \frac{1}{2n(2n+2)}$.

Solution

1. Factor each denominator as $4j(j+1)$.
2. $\frac{1}{4}\left(1 - \frac{1}{n+1}\right) = \frac{n}{4(n+1)}$

Final answer

$$\frac{n}{4(n+1)}$$

Q204 Written

Find the following sum: $\sum_{k=1}^n \frac{1}{\sqrt{k+2} + \sqrt{k+1}}$.

Solution

1. Multiply by the conjugate.
2. $\frac{\sqrt{k+2}-\sqrt{k+1}}{(k+2)-(k+1)} = \sqrt{k+2} - \sqrt{k+1}$
3. $\sum_{k=1}^n (\sqrt{k+2} - \sqrt{k+1}) = \sqrt{n+2} - \sqrt{2}$

Final answer

$$\sqrt{n+2} - \sqrt{2}$$

Worked Solutions

Summation

Q205 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{k+1} + \sqrt{k}}.$

Solution

1. Rationalise with $\sqrt{k+1} - \sqrt{k}.$
2. $\sum_{k=1}^n (\sqrt{k+1} - \sqrt{k}) = \sqrt{n+1} - 1$

Final answer

$$\sqrt{n+1} - 1$$

Q206 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{k} + \sqrt{k-1}}.$

Solution

1. Rationalise: the denominator difference is 1.
2. $\sum_{k=1}^n (\sqrt{k} - \sqrt{k-1}) = \sqrt{n}$

Final answer

$$\sqrt{n}$$

Worked Solutions

Summation

Q207 Written

Find $\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \dots + \frac{1}{\sqrt{n} + \sqrt{n+1}}.$

Solution

1. Use the conjugate to obtain $\sqrt{j+1} - \sqrt{j}.$
2. $\sum_{j=1}^n (\sqrt{j+1} - \sqrt{j}) = \sqrt{n+1} - 1$

Final answer

$$\sqrt{n+1} - 1$$

Q208 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{2k-1} + \sqrt{2k+1}}.$

Solution

1. Rationalise; the square-difference gap is 2.
2. $\frac{1}{2} \sum_{k=1}^n (\sqrt{2k+1} - \sqrt{2k-1}) = \frac{\sqrt{2n+1}-1}{2}$

Final answer

$$\frac{\sqrt{2n+1} - 1}{2}$$

Worked Solutions

Summation

Q209 Written

Find $\frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \cdots + \frac{1}{\sqrt{n+1} + \sqrt{n+2}}.$

Solution

1. Multiply by each conjugate.
2. $\sum_{j=1}^n (\sqrt{j+2} - \sqrt{j+1}) = \sqrt{n+2} - \sqrt{2}$

Final answer

$$\sqrt{n+2} - \sqrt{2}$$

Worked Solutions

Summation

Q210 Challenge

Find the sum, showing your steps: $\sum_{r=1}^{30} \frac{1}{r(r+2)}$.

Solution

1. Use $\frac{1}{r(r+2)} = \frac{1}{2} \left(\frac{1}{r} - \frac{1}{r+2} \right)$.
2. $S = \frac{1}{2} \left(1 + \frac{1}{2} - \frac{1}{31} - \frac{1}{32} \right)$
3. $S = \frac{1425}{1984}$

Final answer

$$\frac{1425}{1984}$$

Worked Solutions

Summation

Q211 Written

Find the following sum $S = 3 \times 1 + 5 \times 3 + 7 \times 3^2 + 9 \times 3^3 + \cdots + (2n + 1) \times 3^{n-1}$.

Solution

1. The arithmetic coefficient is $2k + 1$ and the geometric ratio is $r = 3$.
2. Write S and $3S$, subtract with aligned powers, and simplify: $-2S = -2n \times 3^n$.
3. $S = n \times 3^n$

Final answer

$$n \times 3^n$$

Q212 Written

Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$.

Solution

1. Write the first sum and multiply by the common ratio.
2. Subtract and keep the final shifted term with its correct sign.
3. $3^n n$

Final answer

$$3^n n$$

Worked Solutions

Summation

Q213 Written

Find the sum $\sum_{k=1}^4 (2k + 1) \times (3)^{k-1}$.

Solution

1. Identify the arithmetic coefficient and geometric ratio.
2. Apply (S-rS) over the finite limit ($n = 4$).
3. 324

Final answer

324

Q214 Written

Find the sum $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$.

Solution

1. Write the first sum and multiply by the common ratio.
2. Subtract and keep the final shifted term with its correct sign.
3. $\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}$

Final answer

$$\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}$$

Worked Solutions

Summation

Q215 Written

Find the sum $\sum_{k=1}^6 (2k - 1) \times (4)^{k-1}$.

Solution

1. Identify the arithmetic coefficient and geometric ratio.
2. Apply (S-rS) over the finite limit ($n = 6$).
3. 14109

Final answer

14109

Q216 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer $3^n n$

Worked Solutions

Summation

Q217 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}$$

Q218 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k + 4) \times (2)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$2^n n + 3 \cdot 2^n - 3$$

Worked Solutions

Summation

Q219 Written

Find the following sum $S = 2 \times 1 + 3 \times 3 + 4 \times 3^2 + 5 \times 3^3 + \dots + (n + 1) \times 3^{n-1}$.

Solution

1. Here the coefficient is $k + 1$ and $r = 3$.
2. Subtract $3S$ from S ; the interior coefficients reduce to a geometric sum of 1s.
3. $-2S = \frac{1 - (2n+1) \times 3^n}{2}$
4. $S = \frac{(2n+1) \times 3^n - 1}{4}$

Final answer

$$\frac{(2n + 1) \times 3^n - 1}{4}$$

Q220 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k + 1) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{2 \cdot 3^n n + 3^n - 1}{4}$$

Worked Solutions

Summation

Q221 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k-2) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}$$

Q222 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k-3) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$3^n n - 2 \cdot 3^n + 2$$

Worked Solutions

Summation

Q223 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k + 1) \times (-2)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$-\frac{6(-2)^n n + 5(-2)^n - 5}{9}$$

Worked Solutions

Summation

Q224 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k+2) \times (-3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$-\frac{4(-3)^n n + 9(-3)^n - 9}{16}$$

Worked Solutions

Summation

Q225 Written

Find the following sum $S = 1 \times 1 + 2 \times 2 + 3 \times 2^2 + 4 \times 2^3 + \cdots + n \times 2^{n-1}$.

Solution

1. Use $a_k = k$ and $r = 2$.
2. Applying $S - rS$ gives $-S = 1 - (n-1) \times 2^n$.
3. $S = (n-1) \times 2^n + 1$

Final answer

$$(n-1) \times 2^n + 1$$

Q226 Written

Find the following sum $S = 1 \times 3 + 3 \times 3^2 + 5 \times 3^3 + 7 \times 3^4 + \cdots + (2n-1) \times 3^n$.

Solution

1. Factor out the leading 3, leaving $\sum_{k=1}^n (2k-1)3^{k-1}$.
2. The $S - rS$ subtraction gives $\sum_{k=1}^n (2k-1)3^{k-1} = (n-1)3^n + 1$.
3. $S = (n-1) \times 3^{n+1} + 3$

Final answer

$$(n-1) \times 3^{n+1} + 3$$

Worked Solutions

Summation

Q227 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (3k) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}$$

Worked Solutions

Summation

Q228 Challenge

Find the sum of the partial sums $5 + (5 + 9) + (5 + 9 + 13) + \cdots + [5 + 9 + \cdots + (4k + 1)]$.

Solution

1. The j -th inner arithmetic sum is $5 + 9 + \cdots + (4j + 1) = j(2j + 3)$.
2. Sum the partial sums: $\sum_{j=1}^k (2j^2 + 3j)$.
3. $\frac{k(k+1)(4k+11)}{6}$

Final answer

$$\frac{k(k+1)(4k+11)}{6}$$

Worked Solutions

Summation

Q229 Written

For the sequence 3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, ..., find the k -th term.

Solution

1. The inner terms form an arithmetic sequence with first term 3 and common difference 2.
2. $u_i = 2i + 1$, $a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(3) + (k - 1)(2)] = k(k + 2)$

Final answer

$$a_k = k(k + 2)$$

Q230 Written

For the sequence 1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, ..., find the k -th term.

Solution

1. The inner terms form an arithmetic sequence with first term 1 and common difference 2.
2. $u_i = 2i - 1$, $a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(1) + (k - 1)(2)] = k^2$

Final answer

$$a_k = k^2$$

Worked Solutions

Summation

Q231 Written

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, **find the k -th term.**

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 2.
2. $u_i = 2i, \quad a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(2) + (k-1)(2)] = k(k+1)$

Final answer

$$a_k = k(k+1)$$

Q232 Written

For the sequence $1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, \dots$, **find the k -th term.**

Solution

1. The inner terms form an arithmetic sequence with first term 1 and common difference 4.
2. $u_i = 4i - 3, \quad a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(1) + (k-1)(4)] = k(2k-1)$

Final answer

$$a_k = k(2k-1)$$

Worked Solutions

Summation

Q233 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, what is a_k :**Solution**

1. The inner terms form an arithmetic sequence with first term 2 and common difference 2.

2. $u_i = 2i, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(2) + (k-1)(2)] = k(k+1)$

Correct option: A**Final answer**

$$k(k+1)$$

Q234 MCQ

Select the correct option: For the cumulative sequence $3, 3 + 6, 3 + 6 + 9, 3 + 6 + 9 + 12, \dots$, what is a_k :**Solution**

1. The inner terms form an arithmetic sequence with first term 3 and common difference 3.

2. $u_i = 3i, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(3) + (k-1)(3)] = \frac{3k(k+1)}{2}$

Correct option: A**Final answer**

$$\frac{3k(k+1)}{2}$$

Worked Solutions

Summation

Q235 MCQ

Select the correct option: For the cumulative sequence 4, 4 + 6, 4 + 6 + 8, 4 + 6 + 8 + 10, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 4 and common difference 2.

2. $u_i = 2i + 2, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(4) + (k-1)(2)] = k(k+3)$

Correct option: A

Final answer

$$k(k+3)$$

Q236 MCQ

Select the correct option: For the cumulative sequence 2, 2 + 6, 2 + 6 + 10, 2 + 6 + 10 + 14, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 4.

2. $u_i = 4i - 2, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(2) + (k-1)(4)] = 2k^2$

Correct option: A

Final answer

$$2k^2$$

Worked Solutions

Summation

Q237 MCQ

Select the correct option: For the cumulative sequence 5, 5 + 8, 5 + 8 + 11, 5 + 8 + 11 + 14, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 5 and common difference 3.

$$2. u_i = 3i + 2, \quad a_k = \sum_{i=1}^k u_i$$

$$3. a_k = \frac{k}{2} [2(5) + (k-1)(3)] = \frac{k(3k+7)}{2}$$

Correct option: A

Final answer

$$\frac{k(3k+7)}{2}$$

Q238 MCQ

Select the correct option: For the cumulative sequence 3, 3 + 8, 3 + 8 + 13, 3 + 8 + 13 + 18, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 3 and common difference 5.

$$2. u_i = 5i - 2, \quad a_k = \sum_{i=1}^k u_i$$

$$3. a_k = \frac{k}{2} [2(3) + (k-1)(5)] = \frac{k(5k+1)}{2}$$

Correct option: A

Final answer

$$\frac{k(5k+1)}{2}$$

Worked Solutions

Summation

Q239 MCQ

Select the correct option: For the cumulative sequence 6, 6 + 8, 6 + 8 + 10, 6 + 8 + 10 + 12, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 6 and common difference 2.

2. $u_i = 2i + 4, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(6) + (k-1)(2)] = k(k+5)$

Correct option: A

Final answer

$$k(k+5)$$

Q240 MCQ

Select the correct option: For the cumulative sequence 4, 4 + 8, 4 + 8 + 12, 4 + 8 + 12 + 16, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 4 and common difference 4.

2. $u_i = 4i, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(4) + (k-1)(4)] = 2k(k+1)$

Correct option: A

Final answer

$$2k(k+1)$$

Worked Solutions

Summation

Q241 Written

For the sequence 3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, ..., find the sum from the first term to the n -th term.

Solution

1. First collapse the inner arithmetic sum: $a_k = k(k + 2)$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(2n+7)}{6}$

Final answer

$$S_n = \frac{n(n+1)(2n+7)}{6}$$

Q242 Written

For the sequence 1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, ..., find the sum from the first term to the n -th term.

Solution

1. First collapse the inner arithmetic sum: $a_k = k^2$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(2n+1)}{6}$

Final answer

$$S_n = \frac{n(n+1)(2n+1)}{6}$$

Worked Solutions

Summation

Q243 Written

For the sequence $1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, \dots$, **find the sum from the first term to the n -th term.**

Solution

1. First collapse the inner arithmetic sum: $a_k = k(2k - 1)$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(4n-1)}{6}$

Final answer

$$S_n = \frac{n(n+1)(4n-1)}{6}$$

Q244 Written

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, **find the sum from the first term to the n -th term.**

Solution

1. First collapse the inner arithmetic sum: $a_k = k(k + 1)$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(n+2)}{3}$

Final answer

$$S_n = \frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Summation

Q245 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 5, 2 + 5 + 8, 2 + 5 + 8 + 11, \dots$, find a_5 :

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 3.

2. $u_i = 3i - 1, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(2) + (k - 1)(3)] = \frac{k(3k+1)}{2}$

4. $a_5 = 40$

Correct option: A

Final answer

40

Worked Solutions

Summation

Q246 Challenge

The sum of the first n terms of $\{a_n\}$ is $S_n = 2n^2 + 5n$. Find the general term.

Solution

1. $a_1 = S_1 = 7$
2. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
3. $a_n = (2n^2 + 5n) - [2(n-1)^2 + 5(n-1)] = 4n + 3$
4. The formula also gives $a_1 = 7$, so it is valid for every positive integer n .

Final answer

$$a_n = 4n + 3$$

Worked Solutions

Summation

Q247 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 3)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 4)(n - 1)$
3. $a_n = 2(n - 2)$
4. $a_1 = S_1 = -2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2(n - 2)$$

Worked Solutions

Summation

Q248 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(2n + 5)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(2n+3)$
3. $a_n = 4n + 3$
4. $a_1 = S_1 = 7$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 4n + 3$$

Worked Solutions

Summation

Q249 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)^2$
3. $a_n = 2n - 1$
4. $a_1 = S_1 = 1$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2n - 1$$

Worked Solutions

Summation

Q250 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 4)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 5)(n - 1)$
3. $a_n = 2n - 5$
4. $a_1 = S_1 = -3$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2n - 5$$

Worked Solutions

Summation

Q251 MCQ

Select the correct option: If $S_n = n(n + 2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 1)(n + 1)$
3. $a_n = 2n + 1$
4. $a_1 = S_1 = 3$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 2n + 1$$

Worked Solutions

Summation

Q252 MCQ

Select the correct option: If $S_n = 2n(n + 2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2(n-1)(n+1)$
3. $a_n = 2(2n+1)$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 2(2n + 1)$$

Worked Solutions

Summation

Q253 MCQ

Select the correct option: If $S_n = 3n(n - 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 3(n - 2)(n - 1)$
3. $a_n = 6(n - 1)$
4. $a_1 = S_1 = 0$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: B

Final answer

$$a_n = 6(n - 1)$$

Worked Solutions

Summation

Q254 MCQ

Select the correct option: If $S_n = 2n(2n - 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2(n-1)(2n-3)$
3. $a_n = 2(4n-3)$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: A

Final answer

$$a_n = 2(4n - 3)$$

Worked Solutions

Summation

Q255 MCQ

Select the correct option: If $S_n = 5n(n + 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 5n(n - 1)$
3. $a_n = 10n$
4. $a_1 = S_1 = 10$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 10n$$

Worked Solutions

Summation

Q256 MCQ

Select the correct option: If $S_n = n(2n + 3)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(2n+1)$
3. $a_n = 4n + 1$
4. $a_1 = S_1 = 5$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 4n + 1$$

Worked Solutions

Summation

Q257 MCQ

Select the correct option: If $S_n = n(3n + 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(3n-2)$
3. $a_n = 2(3n-1)$
4. $a_1 = S_1 = 4$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 2(3n - 1)$$

Worked Solutions

Summation

Q258 MCQ

Select the correct option: If $S_n = n(n^2 + 2n + 3)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(n^2 + 2)$
3. $a_n = 3n^2 + n + 2$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 3n^2 + n + 2$$

Worked Solutions

Summation

Q259 MCQ

Select the correct option: If $S_n = n(n+1)(2n-1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n(n-1)(2n-3)$
3. $a_n = 2n(3n-2)$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: B

Final answer

$$a_n = 2n(3n-2)$$

Worked Solutions

Summation

Q260 MCQ

Select the correct option: If $S_n = n(3n^2 - 2n + 4)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(3n^2 - 8n + 9)$
3. $a_n = 9n^2 - 13n + 9$
4. $a_1 = S_1 = 5$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: A

Final answer

$$a_n = 9n^2 - 13n + 9$$

Worked Solutions

Summation

Q261 MCQ

Select the correct option: If $S_n = n(n^2 - n + 2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(n^2 - 3n + 4)$
3. $a_n = 3n^2 - 5n + 4$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 3n^2 - 5n + 4$$

Worked Solutions

Summation

Q262 MCQ

Select the correct option: If $S_n = n(n+1)(2n+1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n(n-1)(2n-1)$
3. $a_n = 6n^2$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 6n^2$$

Worked Solutions

Summation

Q263 MCQ

Select the correct option: If $S_n = n(n+1)(n+2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n(n-1)(n+1)$
3. $a_n = 3n(n+1)$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: B

Final answer

$$a_n = 3n(n+1)$$

Worked Solutions

Summation

Q264 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^3 + 2n + 6$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n^3 - 3n^2 + 5n + 3$
3. $a_n = 3(n^2 - n + 1)$
4. $a_1 = S_1 = 9$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 9, \quad a_n = 3(n^2 - n + 1) \quad (n \geq 2)$$

Worked Solutions

Summation

Q265 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2 + 2n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n^2 - 2$
3. $a_n = 2n + 1$
4. $a_1 = S_1 = 2$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 2, \quad a_n = 2n + 1 \quad (n \geq 2)$$

Worked Solutions

Summation

Q266 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = (n-1)(n^2 + n + 1)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-2)(n^2 - n + 1)$
3. $a_n = 3n^2 - 3n + 1$
4. $a_1 = S_1 = 0$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 0, \quad a_n = 3n^2 - 3n + 1 \quad (n \geq 2)$$

Worked Solutions

Summation

Q267 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{5^n - 5}{5}$
3. $a_n = 4 \cdot 5^{n-1}$
4. $a_1 = S_1 = 4$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 4 \cdot 5^{n-1}$$

Worked Solutions

Summation

Q268 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 3^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{3^n - 3}{3}$
3. $a_n = 2 \cdot 3^{n-1}$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2 \cdot 3^{n-1}$$

Worked Solutions

Summation

Q269 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 2^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{2^n - 2}{2}$
3. $a_n = 2^{n-1}$
4. $a_1 = S_1 = 1$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2^{n-1}$$

Worked Solutions

Summation

Q270 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 7^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{7^n - 7}{7}$
3. $a_n = 6 \cdot 7^{n-1}$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 6 \cdot 7^{n-1}$$

Worked Solutions

Summation

Q271 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 2$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{5^n - 10}{5}$
3. $a_n = 4 \cdot 5^{n-1}$
4. $a_1 = S_1 = 3$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 3, \quad a_n = 4 \cdot 5^{n-1} \quad (n \geq 2)$$

Worked Solutions

Summation

Q272 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 4 \cdot 2^n - 3$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 4 \cdot 2^{n-1} - 3$
3. $a_n = 2^{n+1}$
4. $a_1 = S_1 = 5$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 5, \quad a_n = 2^{n+1} \quad (n \geq 2)$$

Worked Solutions

Summation

Q273 MCQ

Select the correct option: If $S_n = n(n + 4)$, find a_5 :**Solution**

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 1)(n + 3)$
3. $a_n = 2n + 3$
4. $a_1 = S_1 = 5$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .
6. $a_5 = 13$

Correct option: B**Final answer**

13

Worked Solutions

Summation

Q274 Challenge

Find the general term of the sequence $\{a_n\}$ where S_n is given by $S_n = 3 \cdot 3^n - 2n + 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 3^n - 2n + 3$
3. $a_n = 2(3^n - 1)$
4. $a_1 = S_1 = 8$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 8, \quad a_n = 2(3^n - 1) \quad (n \geq 2)$$

Worked Solutions

Summation

Quiz MCQ 1 **Concept Quiz · MCQ**

Select the correct option: Expand $\sum_{k=1}^4 (3k - 1)$ into a sum of terms:

Solution

1. Substitute $k = 1, 2, 3, 4$.

Correct option: A

Final answer

$$2 + 5 + 8 + 11$$

Quiz MCQ 2 **Concept Quiz · MCQ**

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:

Solution

1. Expand $k(k^2 - 1) = k^3 - k$.

Correct option: A

Final answer

$$\frac{n(n+1)(n-1)(n+2)}{4}$$

Worked Solutions

Summation

Quiz MCQ 3 **Concept Quiz · MCQ**

Select the correct option: Find the sum $\sum_{k=1}^n k \times 2^{k-1}$:

Solution

1. Write S and rS, subtract, and isolate S.

Correct option: A

Final answer

$$(n - 1) \times 2^n + 1$$

Worked Solutions

Summation

Quiz MCQ 4 Concept Quiz · MCQ

Select the correct option: If $S_n = 2^n - 3$, find a_6 :

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{2^n - 6}{2}$
3. $a_n = 2^{n-1}$
4. $a_1 = S_1 = -1$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.
6. $a_6 = 32$

Correct option: C

Final answer

32

Worked Solutions

Summation

Quiz WRITTEN 5 Concept Quiz · Written

Expand $\sum_{k=1}^5 (k^2 - 2)$ into a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.

Final answer

$$-1 + 2 + 7 + 14 + 23$$

Quiz WRITTEN 6 Concept Quiz · Written

Find the sum $\sum_{k=1}^n k^2(k + 5)$.

Solution

1. Expand into $(k^3 + 5k^2)$.
2. Use the standard power sums and factor.

Final answer

$$\frac{n(n+1)(3n^2 + 23n + 10)}{12}$$

Worked Solutions

Summation

Quiz WRITTEN 7 Concept Quiz · Written

Find the sum $\sum_{k=1}^n (4k - 1) \times 3^k$.

Solution

1. Show the S-rS alignment (or the inner arithmetic sum for the challenge).
2. $\frac{3(4 \cdot 3^n n - 3 \cdot 3^n + 3)}{2}$

Final answer

$$\frac{3(4 \cdot 3^n n - 3 \cdot 3^n + 3)}{2}$$

Worked Solutions

Summation

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term of the sequence $\{a_n\}$ when S_n is $2 \cdot 2^n - 2n + 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2^n - 2n + 3$
3. $a_n = 2^n - 2$
4. $a_1 = S_1 = 3$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 3, \quad a_n = 2^n - 2 \quad (n \geq 2)$$

Answer key

Use the question number shown on each page.

Q001 $2 + 4 + 6 + 8 + 10$

Q002 $3 + 5 + 7 + 9 + 11$

Q003 $3 + 6 + 9 + 12 + 15$

Q004 $2 + 6 + 10 + 14$

Q005 $-1 + 1 + 3 + 5 + 7$

Q006 A

Q007 A

Q008 A

Q009 A

Q010 A

Q011 A

Q012 A

Q013 A

Q014 $-2 + 1 + 6$

Q015 $5 + 14 + 29$

Q016 $1 + 4 + 9 + 16$

Q017 $5 + 11 + 21$

Q018 $-1 + 0 + 3 + 8$

Q019 A

Q020 A

Q021 A

Q022 A

Q023 A

Q024 A

Q025 A

Q026 A

Q027 $3 + 9 + 27 + 81$

Q028 $2 + 2 + 2$

Q029 $2 + 4 + 8 + 16$

Q030 $3 + 3 + 3 + 3 + 3 + 3$

Q031 $3 + 9 + 27$

Q032 $8 + 28 + 128$

Q033 $-1 + 1 + 5 + 13$

Q034 $5 + 5 + 5 + 5$

Q035 $3 + 3 + 3$

Q036 $4 + 4 + 4 + 4 + 4$

Q037 A

Q038 A

Q039 A

Q040 A

Q041 A

Q042 A

Q043 A

Q044 A

Q045 $\sum_{k=1}^n (4k - 1)$

Q046 $\sum_{k=1}^n (3k + 2)$

Q047 $\sum_{k=1}^n (2k + 1)$

Q048 $\sum_{k=1}^n (2k - 1)$

Q049 $\sum_{k=1}^n (7k - 1)$

Q050 $\sum_{k=1}^n (4k + 5)$

Q051 $\sum_{k=1}^n (3k - 2)$

Q052 A

Q053 A

Q054 A

Q055 A

Q056 A

Q057 A

Q058 A

Q059 A

Q060 $\sum_{k=6}^{15} k^2$

Q061 $\sum_{k=4}^{13} k^2$

Q062 $\sum_{k=1}^n (3^k - 2)$

Q063 $\sum_{k=1}^n 2^{k-1}$

Q064 $\sum_{k=7}^{21} k^2$

Q065 $\sum_{k=12}^{18} k^2$

Q066 $\sum_{k=10}^{20} k^2$

Q067 A

Q068 A

Q069 A

Q070 A

Q071 A

Q072 A

Q073 A

Q074 $\sum_{k=1}^{10} (3k - 1)$

Q075 $\sum_{k=1}^7 8k$

Q076 $\sum_{k=1}^{12} (6k + 1)$

Q077 $\sum_{k=1}^{24} (5k - 1)$

Q078 $\sum_{k=1}^n (3k - 2)$

Q079 $-3n$

Q080 $3n$

Q081 $6n$

Q082 $-4n$

Q083 n

Q084 A

Q085 A

Q086 A

Q087 A

Q088 A

Q089 A	Q118 $\frac{n(n+1)(2n-5)}{6}$	Q145 A	Q174 62
Q090 A	Q119 $\frac{2n(n+1)(n-1)}{3}$	Q146 A	Q175 60
Q091 A	Q120 $\frac{n(n+1)(n+5)}{3}$	Q147 A	Q176 105
Q092 A	Q121 $\frac{n(n+1)(2n+3)}{2}$	Q148 A	Q177 $\frac{n}{2(n+2)}$
Q093 A	Q122 $\frac{n(n+4)(n+5)}{3}$	Q149 A	Q178 $\frac{n}{n+1}$
Q094 A	Q123 16	Q150 A	Q179 $\frac{n}{3(n+3)}$
Q095 A	Q124 35	Q151 A	Q180 A
Q096 A	Q125 280	Q152 A	Q181 A
Q097 A	Q126 184	Q153 A	Q182 A
Q098 A	Q127 100	Q154 A	Q183 A
Q099 A	Q128 245	Q155 A	Q184 A
Q100 $\frac{n(5n+13)}{2}$	Q129 $\frac{n(n+1)(n+2)(3n+1)}{12}$	Q156 A	Q185 A
Q101 $n(n-2)$	Q130 $\frac{n(n+1)(3n^2-5n-4)}{12}$	Q157 A	Q186 A
Q102 $n(2n+1)$	Q131 $\frac{n(n+1)(3n^2+11n+4)}{12}$	Q158 A	Q187 A
Q103 $\frac{n(3n+7)}{2}$	Q132 $\frac{n(n+1)(n^2-3n-2)}{4}$	Q159 $3(3^n-1)$	Q188 A
Q104 $3n^2$	Q133 A	Q160 $4(2^n-1)$	Q189 $\frac{n}{2(3n+2)}$
Q105 A	Q134 A	Q161 $2(2^n-1)$	Q190 $\frac{n}{2n+1}$
Q106 A	Q135 A	Q162 $\frac{1-(-3)^n}{2}$	Q191 $\frac{1}{2}(1+\frac{1}{2}-\frac{1}{n+1}-\frac{1}{n+2})$
Q107 A	Q136 A	Q163 $1-(-2)^n$	Q192 $\frac{n}{3n+1}$
Q108 A	Q137 A	Q164 $\frac{5((-5)^n-1)}{6}$	Q193 $\frac{1}{6}(1+\frac{1}{4}-\frac{1}{3n+1}-\frac{1}{3n+4})$
Q109 $\frac{n(n-1)(2n-1)}{6}$	Q138 A	Q165 $1-(-1)^n$	Q194 A
Q110 A	Q139 A	Q166 $2(1-(-3)^n)$	Q195 A
Q111 A	Q140 A	Q167 $\frac{1-(-2)^n}{3}$	Q196 A
Q112 A	Q141 $\frac{n(n+1)(n-1)(n+2)}{4}$	Q168 $\frac{3((-3)^n-1)}{4}$	Q197 A
Q113 A	Q142 $\frac{n(n+1)(n^2+n+4)}{4}$	Q169 $\frac{4(4^n-1)}{3}$	Q198 A
Q114 A	Q143 A	Q170 $\frac{4^n-1}{3}$	Q199 A
Q115 $\frac{n(n+1)(10n-1)}{6}$	Q144 A	Q171 $\frac{3^n-1}{2}$	Q200 A
Q116 $\frac{n(n-1)(2n+1)}{2}$		Q172 -5	Q201 A
Q117 $\frac{n(n+1)(4n+5)}{6}$		Q173 39	Q202 $\frac{1}{2}(\frac{1}{4}+\frac{1}{5}-\frac{1}{n+4}-\frac{1}{n+5})$

$$\text{Q203 } \frac{n}{4(n+1)}$$

$$\text{Q204 } \sqrt{n+2} - \sqrt{2}$$

$$\text{Q205 } \sqrt{n+1} - 1$$

$$\text{Q206 } \sqrt{n}$$

$$\text{Q207 } \sqrt{n+1} - 1$$

$$\text{Q208 } \frac{\sqrt{2n+1} - 1}{2}$$

$$\text{Q209 } \sqrt{n+2} - \sqrt{2}$$

$$\text{Q210 } \frac{1425}{1984}$$

$$\text{Q211 } n \times 3^n$$

$$\text{Q212 } 3^n n$$

$$\text{Q213 } 324$$

$$\text{Q214 } \frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n} + 5}{9}$$

$$\text{Q215 } 14109$$

$$\text{Q216 } A$$

$$\text{Q217 } A$$

$$\text{Q218 } A$$

$$\text{Q219 } \frac{(2n+1) \times 3^n - 1}{4}$$

$$\text{Q220 } A$$

$$\text{Q221 } A$$

$$\text{Q222 } A$$

$$\text{Q223 } A$$

$$\text{Q224 } A$$

$$\text{Q225 } (n-1) \times 2^n + 1$$

$$\text{Q226 } (n-1) \times 3^{n+1} + 3$$

$$\text{Q227 } A$$

$$\text{Q228 } \frac{k(k+1)(4k+11)}{6}$$

$$\text{Q229 } a_k = k(k+2)$$

$$\text{Q230 } a_k = k^2$$

$$\text{Q231 } a_k = k(k+1)$$

$$\text{Q232 } a_k = k(2k-1)$$

$$\text{Q233 } A$$

$$\text{Q234 } A$$

$$\text{Q235 } A$$

$$\text{Q236 } A$$

$$\text{Q237 } A$$

$$\text{Q238 } A$$

$$\text{Q239 } A$$

$$\text{Q240 } A$$

$$\text{Q241 } S_n = \frac{n(n+1)(2n+7)}{6}$$

$$\text{Q242 } S_n = \frac{n(n+1)(2n+1)}{6}$$

$$\text{Q243 } S_n = \frac{n(n+1)(4n-1)}{6}$$

$$\text{Q244 } S_n = \frac{n(n+1)(n+2)}{3}$$

$$\text{Q245 } A$$

$$\text{Q246 } a_n = 4n + 3$$

$$\text{Q247 } a_n = 2(n-2)$$

$$\text{Q248 } a_n = 4n + 3$$

$$\text{Q249 } a_n = 2n - 1$$

$$\text{Q250 } a_n = 2n - 5$$

$$\text{Q251 } D$$

$$\text{Q252 } C$$

$$\text{Q253 } B$$

$$\text{Q254 } A$$

$$\text{Q255 } D$$

$$\text{Q256 } C$$

$$\text{Q257 } C$$

$$\text{Q258 } C$$

$$\text{Q259 } B$$

$$\text{Q260 } A$$

$$\text{Q261 } D$$

$$\text{Q262 } D$$

$$\text{Q263 } B$$

$$\text{Q264 } a_1 = 9, \quad a_n = 3(n^2 - n + 1) \quad (n \geq 2)$$

$$\text{Q265 } a_1 = 2, \quad a_n = 2n + 1 \quad (n \geq 2)$$

$$\text{Q266 } a_1 = 0, \quad a_n = 3n^2 - 3n + 1 \quad (n \geq 2)$$

$$\text{Q267 } a_n = 4 \cdot 5^{n-1}$$

$$\text{Q268 } a_n = 2 \cdot 3^{n-1}$$

$$\text{Q269 } a_n = 2^{n-1}$$

$$\text{Q270 } a_n = 6 \cdot 7^{n-1}$$

$$\text{Q271 } a_1 = 3, \quad a_n = 4 \cdot 5^{n-1} \quad (n \geq 2)$$

$$\text{Q272 } a_1 = 5, \quad a_n = 2^{n+1} \quad (n \geq 2)$$

$$\text{Q273 } B$$

$$\text{Q274 } a_1 = 8, \quad a_n = 2(3^n - 1) \quad (n \geq 2)$$

$$\text{Quiz MCQ 1 } A$$

$$\text{Quiz MCQ 2 } A$$

$$\text{Quiz MCQ 3 } A$$

$$\text{Quiz MCQ 4 } C$$

$$\text{Quiz WRITTEN 5 } -1 + 2 + 7 + 14 + 23$$

$$\text{Quiz WRITTEN 6 } \frac{n(n+1)(3n^2 + 23n + 10)}{12}$$

$$\text{Quiz WRITTEN 7 } \frac{3(4 \cdot 3^n n - 3 \cdot 3^n + 3)}{2}$$

$$\text{Quiz WRITTEN 8 } a_1 = 3, \quad a_n = 2^n - 2 \quad (n \geq 2)$$